

WINDOWS SERVER ROLES & FEATURES

ESSENTIAL SERVER ROLES

Active Directory Domain Services (AD DS)

- Centralized authentication & authorization
- User/computer account management
- Group Policy implementation
- DNS integration & forest/domain structure

DNS Server

- Name resolution (FQDN to IP)
- Forward/reverse lookup zones
- Active Directory integration
- Conditional forwarding & recursion

NETWORK SERVICES

Role	Purpose
DHCP	IP address assignment
DNS	Name resolution
WINS	NetBIOS name resolution
RRAS	VPN/NAT/Routing
NPS	RADIUS authentication
Network Controller	SDN management

FILE & STORAGE

Role	Purpose
File Services	File sharing/management
Storage Services	iSCSI/storage management
DFS	Distributed file system
FSRM	File Server Resource Mgr
Storage Replica	Block-level replication
Data Deduplication	Storage optimization

WEB & APPLICATION

Role	Purpose
IIS	Web server/hosting
Application Server	.NET Framework apps
WDS	Windows deployment
WSUS	Windows update management
Remote Desktop	Terminal services
Volume Activation	KMS/MAK licensing

ACTIVE DIRECTORY ROLES

- **AD DS:** Domain Services (core directory)
- **AD CS:** Certificate Services (PKI)
- **AD FS:** Federation Services (SSO)
- **AD LDS:** Lightweight Directory Services
- **AD RMS:** Rights Management Services

Domain Controller Roles (FSMO):

- Schema Master (forest-wide)
- Domain Naming Master (forest-wide)
- PDC Emulator (domain-wide)
- RID Master (domain-wide)
- Infrastructure Master (domain-wide)

VIRTUALIZATION & CONTAINERS

- **Hyper-V:** Virtual machine hosting
- **Containers:** Windows/Docker containers
- **Host Guardian Service:** Shielded VM security
- **Storage Spaces Direct:** Software-defined storage

Hyper-V Features:

- Live Migration • Failover Clustering
- Replica • Shielded VMs
- Virtual Switch • Integration Services
- Checkpoints • Dynamic Memory

SECURITY ROLES

- **NPS:** RADIUS/802.1X auth
- **AD CS:** Certificate authority
- **AD FS:** Claims-based auth
- **Windows Defender:** Antivirus
- **Device Health Attestation:** Device compliance
- **Host Guardian:** Guarded fabric

PRINT & FAX

- **Print Services:** Centralized printing
- **Print Server:** Driver management
- **LPD Service:** Unix/Linux printing
- **Internet Printing:** Web-based printing
- **Fax Server:** Network fax services

MULTIMEDIA & MEDIA

- **Media Services:** Streaming media
- **Windows Media Server:** Media streaming
- **MultiPoint Services:** Multiple users on one PC
- **Remote Desktop Services:** Terminal server

COMMON FEATURES

Feature	Description
.NET Framework	Application framework
BitLocker	Drive encryption
Failover Clustering	High availability
Network Load Balancing	Load distribution
PowerShell ISE	Scripting environment
SNMP Service	Network monitoring
Telnet Client/Server	Remote terminal
TFTP Client	File transfer
Windows Backup	System backup
WinRM	Remote management

QUICK TIPS

- Some roles require reboot
- Install management tools with roles
- Use least privilege principle
- Document installed roles/features
- Test in lab before production
- Consider Server Core for security
- Use PowerShell for automation